University of Wollongong
School of Computing and Information Technology
CSIT121 Object Oriented Design and Programming
Assignment 2

## Objectives

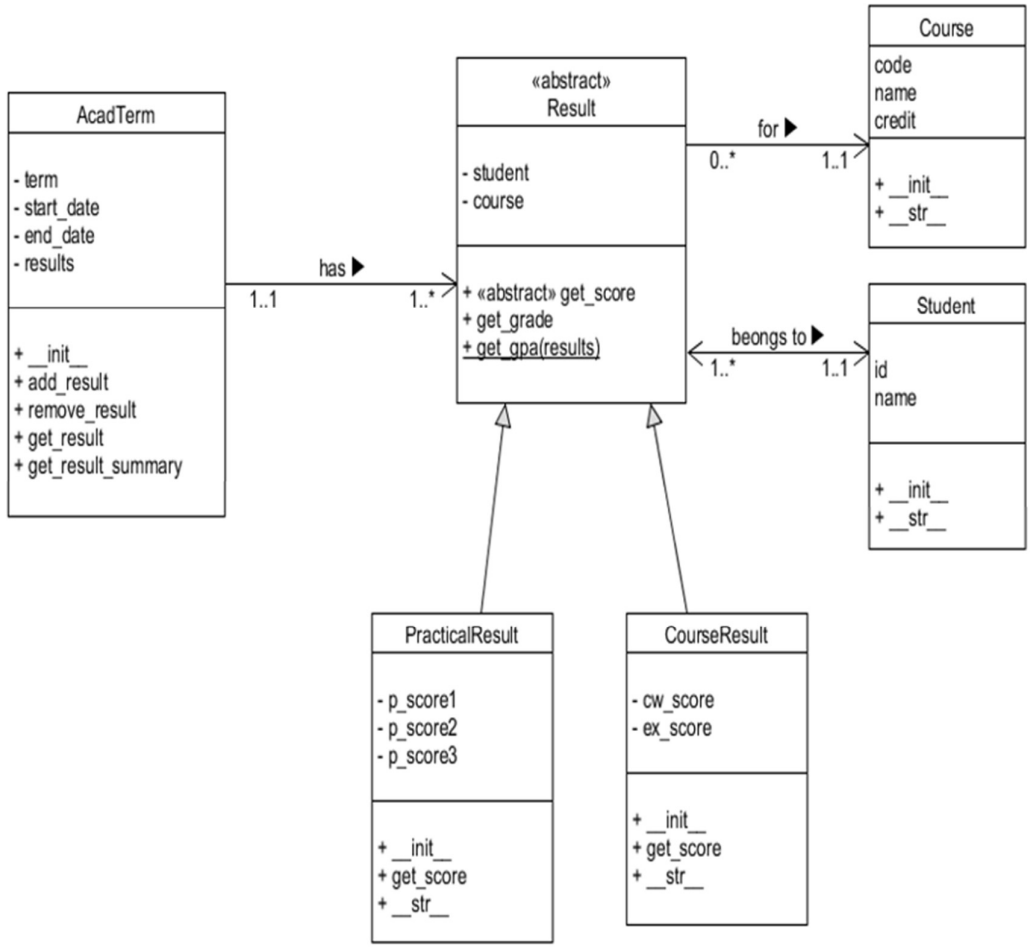
- To apply Object Oriented Design (OOD).
- To apply Object Oriented Programming (OOP) using Python.

## Submission

- Please submit **one text** file containing the Python code (with comments) and the execution results (in text form) to UOW Moodle.
- File name must be in the form of: **TXX_NAME_UOWID.txt** where XX is your tutorial group, NAME is your full name and UOWID is your 7-digit UOW ID number. For example, **T02F_JeffreyTan_8080426.txt**
- Late submission will be penalized 25% per day late. Please refer to UOW Moodle for the assignment due date (in Singapore time).

## Tasks

Write the Python classes to implement the prototype of an Academic Result Management Application depicted in the ***draft*** class diagram.



Class: Course

| **Attribute** | **Description** |
|---|---|
| code | The course code. |
| name | The course name |
| credit | The number of credits of this course |
| **Method** | |
| __init__ | Constructor. |
| __str__ | This method will return a string containing the 3 attributes. |

Class: Student

| **Attribute** | **Description** |
|---|---|
| id | The student id. |
| name | The student's name |
| **Method** | |
| __init__ | Constructor. |
| __str__ | This method will return a string containing the 2 attributes. |

Class: Result

| **Attribute** | **Description** |
|---|---|
| student | The student enrolled in a course. |
| course | The course taken by a student. |
| **Method** | |
| __init__ | Constructor. |
| get_score | This method will compute and return the overall score. It will be overridden. |
| get_grade | This method will compute and return the final grade based on the overall score. Please refer to the grading system described below. |
| __str__ | This method will return a string containing the following:<br>• student id and name<br>• course code<br>• overall score<br>• grade |
| get_gpa | This is a class method that compute and return the GPA for a collection of ***results***. |

Class: PracticalResult

| **Attribute** | **Description** |
|---|---|
| p_score1 | The score of practical work 1. |
| p_score2 | The score of practical work 2. |
| p_score3 | The score of practical work 3. |
| **Method** | |
| __init__ | Constructor. |
| get_score | This method will compute and return the overall score as follow: (p_score1 + p_score2 + p_score3) / 3 |
| __str__ | This method will return a string containing the following:<br>• student id and name<br>• course code<br>• p_score1<br>• p_score2<br>• p_score3<br>• overall score<br>• grade |

Class: CourseResult

| **Attribute** | **Description** |
|---|---|
| cw_score | The score of the course work. |
| ex_score | The score of the exam. |
| **Method** | |
| __init__ | Constructor. |
| get_score | This method will compute and return the overall score as follows: cw_score x 0.3 + ex_score * 0.7 |
| __str__ | This method will return a string containing the following:<br>• student id and name<br>• course code<br>• cw_score<br>• ex_score<br>• overall score<br>• grade |

Class: AcadTerm

| **Attribute** | **Description** |
|---|---|
| term | Term names such as "2024 Q1", "2024 Q2", etc. |
| start_date | First day of the academic term. |
| end_date | Last day of the academic term. |
| results | A collection of ***results*** of the academic term. |
| **Method** | |

| | |
|---|---|
| __init__ | Constructor. |
| add_result | This method will add a result (object of Result subclasses) to the academic term. The method must ensure that the ***same result*** cannot be added multiple times to the same academic term. |
| remove_result | Given the student id and course code, this method will remove the matching result from the academic term. |
| get_result | This method will return all the results for a student (id) or will return the result of a student for a course (if given). |
| get_result_summary | This method will return the summary result academic term. The summary result include:<br>• term<br>• number of passes<br>• number of failures |
| __str__ | The method will return a string containing the following:<br>• term<br>• start_date<br>• end_date |

Grading system

| Grade | Score | Points awarded |
|---|---|---|
| A | >= 95 | 4.0 |
| A- | >= 90 | 3.7 |
| B+ | >= 85 | 3.3 |
| B | >= 80 | 3.0 |
| B- | >= 75 | 2.7 |
| C+ | >= 70 | 2.3 |
| C | >= 65 | 2.0 |
| C- | >= 60 | 1.7 |
| D+ | >= 55 | 1.3 |
| D | >= 50 | 1.0 |
| D- | >= 45 | 0.7 |
| F | < 45 | 0.0 |

Grade point average (GPA)

Assuming a student enrolled in three courses and obtained the following results:

| Grade | Credit | Points awarded |
|---|---|---|
| A | 6 | 4.0 x 6 = 24 |
| B | 6 | 3.0 x 6 = 18 |
| B | 6 | 3.0 x 6 = 18 |
| | Total Credits: 18 | Total Points: 24 + 18 + 18 = 60 |
| | | GPA: 60 / 18 = 3.33 |

You will carry out OOD and OOP as follows:

- You must ***not*** use global variables.
- You must choose an appropriate data type (class) for each attribute.
- You must include appropriate properties and setters (or get and set methods).
- You may decide the appropriate parameter(s) for each method.
- You may include additional attributes and methods for each class.
- You must define a main function with helper functions to thoroughly test the functionalities of the program.
- You must include comments in the program.